IndicLearn — Phase 2 Plan

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IndicLearn already works end to end, but every decision inside it is made by hand-written rules. Phase 2 replaces the four weakest of those rules with small models we train ourselves, so the knowledge map stops showing junk nodes, the quizzes test understanding instead of blanking out a word, and Telugu and Hindi questions find the right passage.

What we have today

A working web app. A student uploads their notes, reads them inside the app, asks doubts to a tutor bot in English, Telugu, Hindi or Sanskrit, takes quizzes, and watches a mind map of the subject fill in as they learn.

Everything inside it runs on rules we wrote by hand. There is no trained model anywhere in the default setup.

Part	How it decides things today
Finding concepts	Counts word patterns and looks for phrases like "X is called Y"
Writing node descriptions	Copies a sentence from the document, if one exists
Drawing connections	Two concepts in the same sentence become "related"
Making quizzes	Deletes a word from a sentence and offers four words
Finding the answer passage	Keyword matching (BM25) and character overlap

This is honest, fast and explainable, and it was the right first version. But rules cannot judge meaning, and that is exactly where it now fails.

What is wrong

We uploaded a real research paper and the map came out unusable. Three of the causes are plain bugs. Three are places where no rule can ever be good enough.

Bugs — we fix these in code, in about one day.

What you see on the map	Why
The centre node is named `faee623705b84365`	The title falls back to the filename, and we save uploads under a random id
"Arti cial", "Classi cation"	PDFs store `fi` as one special character and we never convert it back
Author names and conference names as big nodes	Reference entries like `[1] Smith, G.H., ...` look like numbered headings to our parser
`1-CNN 58.66 56.92 55.87` as a node	A results table row leaked into the concept list

Gaps — these need a trained model.

Problem	Why a rule cannot fix it
Junk like `DCT`, `DWT CWT` becomes a node	Deciding "is this a real concept a student should learn?" is a judgement about meaning, not a pattern
Nodes with no description	If the document never defines the term, there is nothing to copy. Something has to *write* the explanation
Quizzes just blank out a word	A fill-in-the-blank tests memory of a sentence. A real question tests whether you understood the idea

The four models we will train

Each one is small, trained by fine-tuning an existing open model on a dataset we assemble. None of them calls an outside API, so the project rule holds.

M1 — Concept filter. *In plain words: a gatekeeper that decides whether something deserves a circle on the map.* We show it a phrase and the sentence around it, and it answers: real concept, or author name, table number, junk. Base: MuRIL or IndicBERT-v2 (both are Indian-language models from Google and AI4Bharat). Training data: the index and glossary at the back of textbooks give us thousands of confirmed concepts for free, and references and tables give us the negatives.

M2 — Description writer. *In plain words: it writes the one-line explanation under each node when the document does not bother to explain the term.* Base: mT5-small or IndicBART. Training data: the first sentence of Wikipedia articles in English, Telugu and Hindi, plus textbook glossary entries. These are already in the shape "term → definition", which is exactly what we need.

M3 — Relation classifier. *In plain words: it decides which topic you must learn before another one.* We give it two concepts and a sentence; it answers prerequisite, related, or nothing. Base: MuRIL as a pair classifier. Training data: LectureBank, a public dataset of prerequisite links between course topics, plus around 500 pairs we label by hand from Telugu and Hindi notes.

M4 — Quiz generator. *In plain words: it reads a concept and writes a real question with three believable wrong answers.* Base: mT5 or IndicBART. Training data: SciQ, which has about 13,700 science multiple-choice questions where a human wrote the wrong options too, plus RACE and IndicQA. Writing good wrong answers is the hard part, and SciQ is the dataset that teaches it.

M4 is the one that fixes the quiz complaint. M1 and M2 together are what turn the map from a mess into something readable: M1 removes the junk circles, M2 gives every surviving circle a sentence.

The B.Tech knowledge dataset

The models need to know what engineering concepts look like across all four years, not just in one uploaded file. We build one corpus and reuse it for all four models.

Source	What it gives us	Licence
NPTEL lecture transcripts	Indian B.Tech courses, the closest match to our actual users	Free to use, check per-course terms
OpenStax and Wikibooks	Full textbooks in physics, maths and computing	CC-BY, safe to redistribute

Source	What it gives us	Licence
NCERT	The school-level foundation layer	Free
AICTE model curriculum	Subject → unit → topic. This is itself the prerequisite structure M3 needs to learn	Public
Wikipedia in English, Telugu, Hindi	Term → definition pairs for M2	CC-BY-SA

One limit to be clear about. Commercial textbooks such as Oppenheim or Proakis are copyrighted. We can test the app on them privately, but they cannot go into a dataset we publish or hand in, and the report must not claim they did.

Order of work

1. **Fix the four bugs.** About a day. The map stops looking broken, and every later experiment runs on clean input.
2. **Build the corpus.** Collect, clean and split the sources above. Everything downstream waits on this.
3. **Train M4, the quiz generator.** Biggest visible improvement, and SciQ already exists, so there is no labelling to do first.
4. **Train M1 and M2 together.** They share the same corpus and are judged on the same thing: does the map read well?
5. **Train M3.** Needs hand-labelled pairs, so it is slowest to start.
6. **Fine-tune the retriever** (the multilingual search model discussed earlier) if time allows.

Steps 1 and 2 can run in parallel. Training happens on your machine or on Colab or Kaggle — these are small models, not something that needs a cluster.

How we prove it worked

Every model is measured against the rule-based version it replaces. That comparison is the result the report is built on.

Model	Compared against	Measured by	What success looks like
M1 Concept filter	Today's rule-based miner	Precision and recall on a hand-checked set of 200 nodes	Fewer than 1 junk node in 10
M2 Description writer	Copying a sentence from the document	Human rating of 100 descriptions, 1 to 5	Every node has a description, average above 3.5
M3 Relation classifier	Today's co-occurrence rule	F1 on held-out prerequisite pairs	Beats the rule baseline
M4 Quiz generator	Today's fill-in-the-blank	Human rating: is this a fair question, are the wrong answers believable	Majority judged fair

Two honest notes. The mastery scores that colour the personal map are still a heuristic we invented, and no model here validates them. And we have not yet confirmed that every dataset above is currently downloadable and licensed as stated — that check is the first task of step 2.